

EXPERIMENT-11

Performing IP Spoofing Using a GitHub Tool in Kali Linux

AIM

To demonstrate IP spoofing techniques using an open-source GitHub tool in Kali Linux for understanding network-level attack concepts and security vulnerabilities.

Software Requirements

Host Operating System: Windows / Linux / macOS
Virtualization Software: Oracle VirtualBox
Guest Operating System: Kali Linux
Tool Source: GitHub (IP Spoofing Tool-Educational Use)
Programming Language: Python / Bash (as per tool)
System Requirements:
o RAM: Minimum 4 GB (8 GB recommended)
o Disk Space: Minimum 30 GB
o Processor: Intel/AMD with Virtualization support

Introduction

IP spoofing is a technique where an attacker forges the source IP address in network packets to impersonate another system. It is commonly used in attacks such as DDoS, session hijacking, and bypassing IP-based authentication. This experiment demonstrates IP spoofing using a publicly available GitHub tool to understand how attackers manipulate packet headers. Kali Linux provides a safe and controlled environment for ethical network security experimentation.

Procedure

Step 1: Start Kali Linux Virtual Machine

1. Open Oracle VirtualBox.
2. Start Kali Linux
3. Log in using Kali credentials.

Step 2: Open Terminal

1. Launch Terminal in Kali Linux.
2. Update system packages:
3. sudo apt update

Step 3: Clone IP Spoofing Tool from GitHub

1. Clone the repository:
2. git clone https://github.com/<username>/<ip-spoofing-tool>
3. Navigate to the tool directory:
4. cd ip-spoofing-tool

Step 4: Install Dependencies

1. Install required dependencies:
2. sudo apt install python?
3. pip3 install - requirements.txt

Step 5: Configure the Tool

1. Open the script file:
2. nano spoof.py
3. Configure:
4. Target IP address
5. Spoofed source IP address
Network interface

Step 6: Execute IP Spoofing

1. Run the tool with root privileges:
2. sudo python3 spoof.py
3. Observe packet transmission with spoofed IP addresses.

Step 7: Monitor Network Traffic (Optional)

1. Use Wireshark or tcpdump:
2. sudo tcpdump
3. Verify spoofed source IP packets.

OUTPUT:

Packets are sent with forged source IP addresses.

Network traffic reflects spoofed IP information.

Demonstration successfully completed

```
(kali㉿kali)-[~]
└─$ sudo apt install python3-pip -y
Upgrading:
  python3-pip

Summary:
  Upgrading: 1, Installing: 0, Removing: 0, Not Upgrading: 1898
  Download size: 1,390 kB
  Freed space: 47.1 kB

Get:1 http://http.kali.org/kali kali-rolling/main amd64 python3-pip all 26.0.1+dfsg-1 [1,390 kB]
Fetched 1,390 kB in 1s (1,895 kB/s)
(Reading database ... 422160 files and directories currently installed.)
Preparing to unpack ... /python3-pip_26.0.1+dfsg-1_all.deb ...
Unpacking python3-pip (26.0.1+dfsg-1) over (25.2+dfsg-1) ...
Setting up python3-pip (26.0.1+dfsg-1) ...
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for kali-menu (2025.4.3) ...

(kali㉿kali)-[~]
└─$ nano spoof.py

(kali㉿kali)-[~]
└─$ sudo python3 spoof.py
[+] Sending spoofed packet ...
.....
Sent 5 packets.
[+] Packet transmission completed

(kali㉿kali)-[~]
└─$ sudo tcpdump -n icmp
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
```

Result

IP spoofing was successfully demonstrated using a GitHub-based tool in Kali Linux.

The experiment illustrates how attackers can manipulate IP headers and highlights the need for robust network security mechanisms